

ITS Optimization & Connected Vehicles

Symposium on AI for DOTs
August 19, 2026

Donovan Guin
Public Transportation

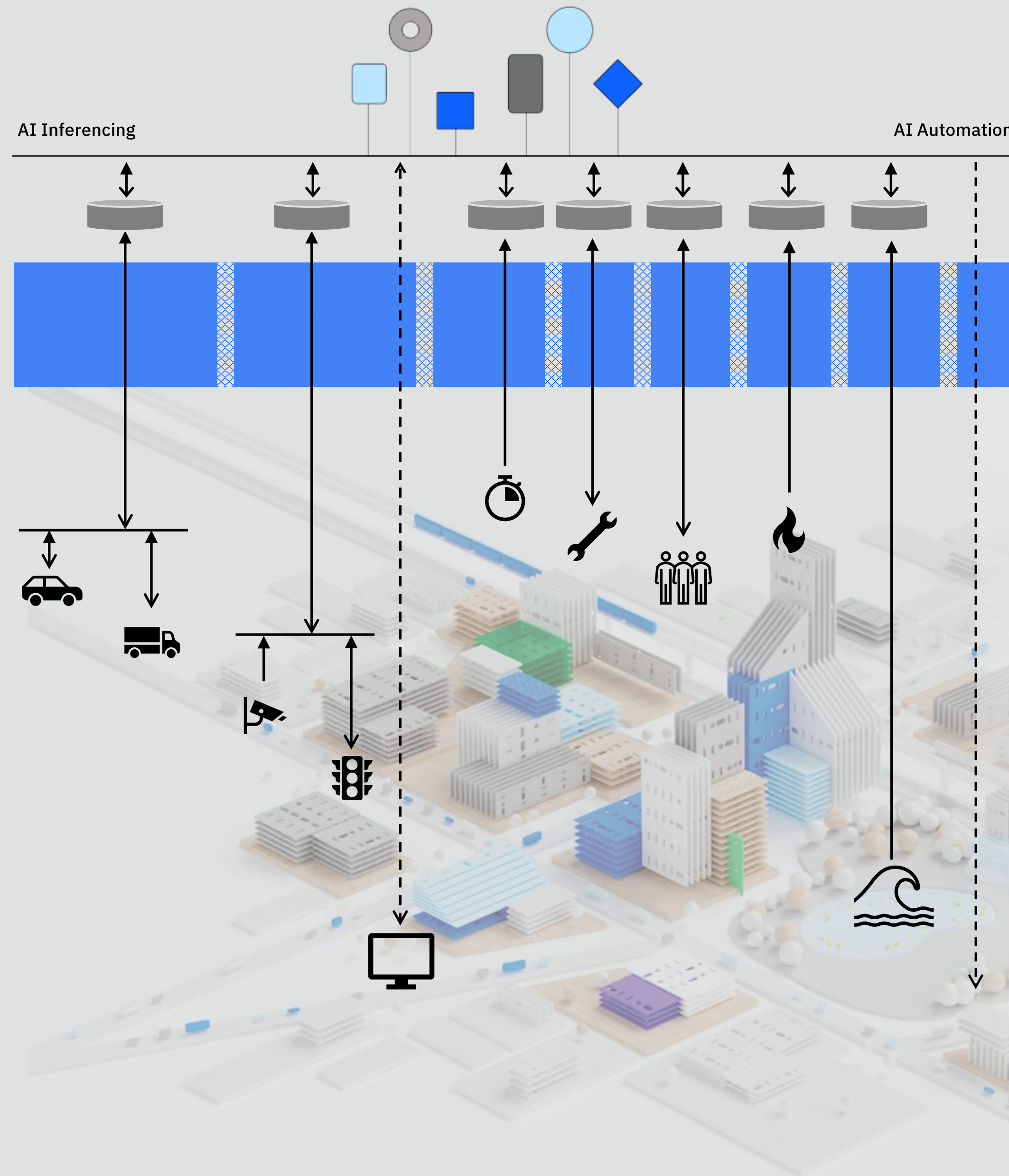


Let's create...

A platform that both commercial and public authorities can use to improve traffic and safety, together.

From siloed systems to an integrated orchestration layer where holistic insights can be turned to action.

No new roadside equipment required.



IBM
Intelligent Mobility Cloud

- **Common integration service** across parties, including cloud and edge computing
- **Common data service** to feed AI agents for priority cases – traffic, safety, asset management
- **Intelligent automation** that can enable **real-time action**



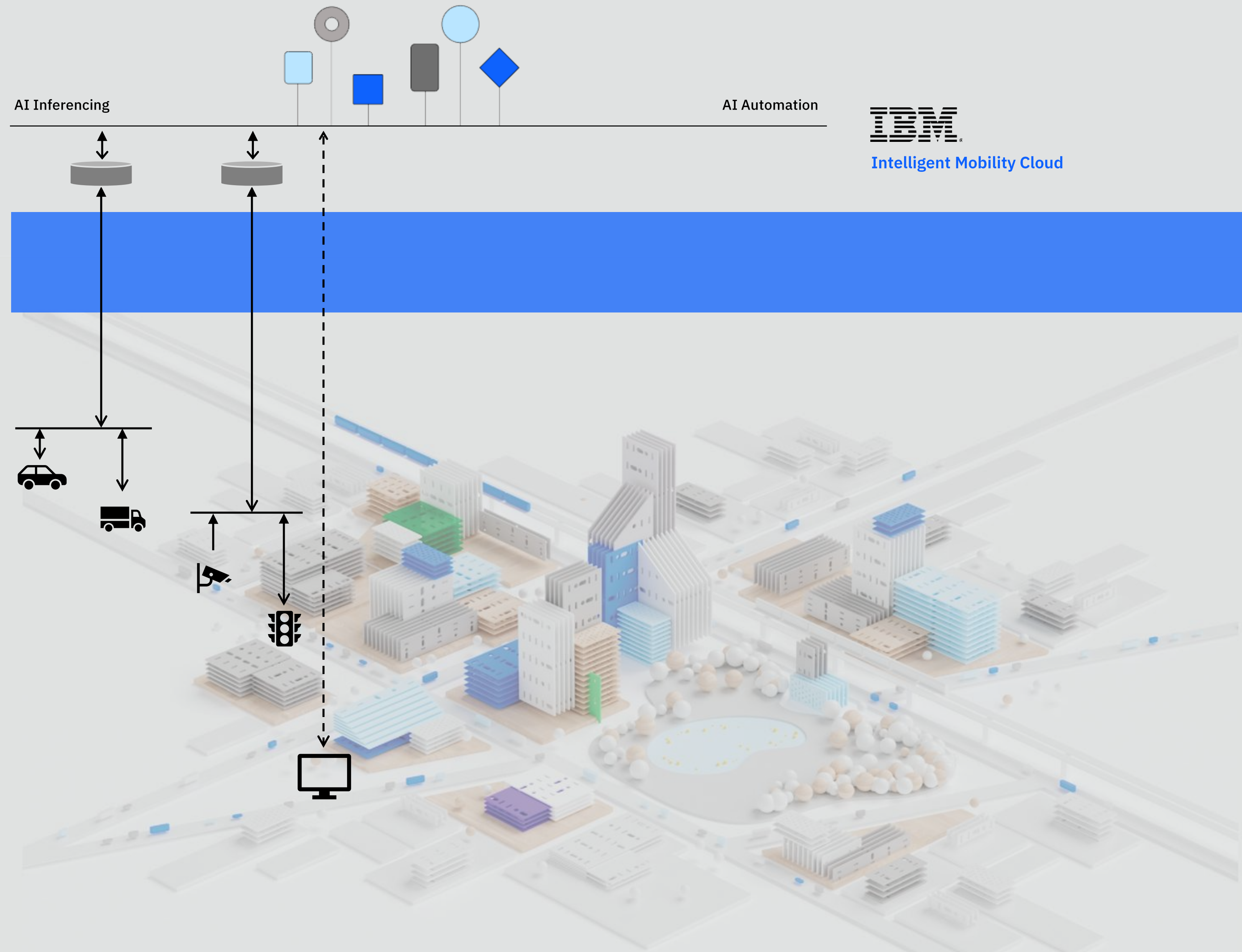
BEST
SESSION
AWARD

For SISO6: Emerging Technologies for Transportation System Management and Operations: A Global Scan

Traffic

Demand sensing can be used to adjust demand fulfillment.

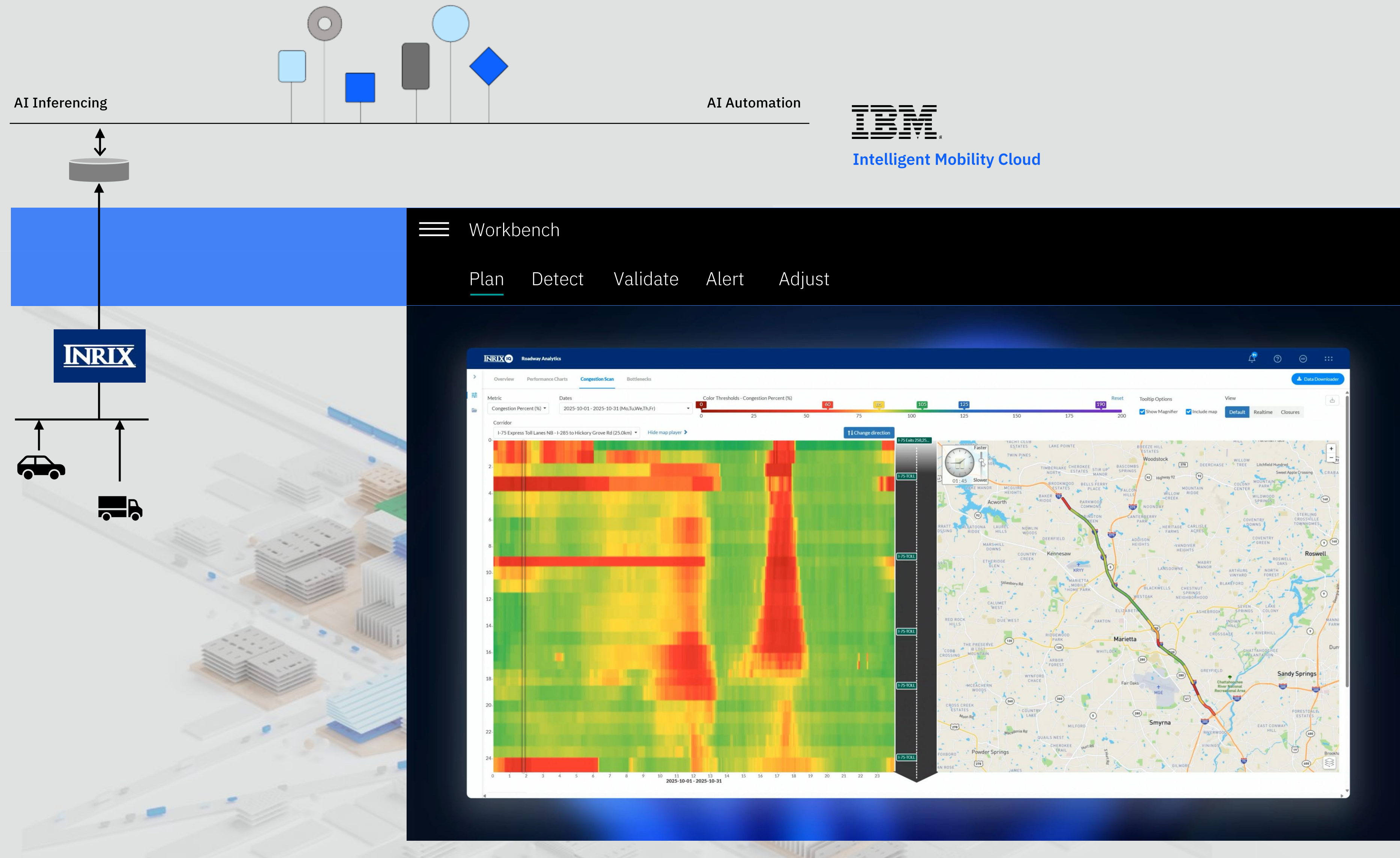
1. Baseline playbooks are tuned based on probe vehicle data
2. Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. Alerts are validated automatically using visual analytics from the camera system already installed
4. Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume



Traffic

Demand sensing can be used to adjust demand fulfillment.

1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume

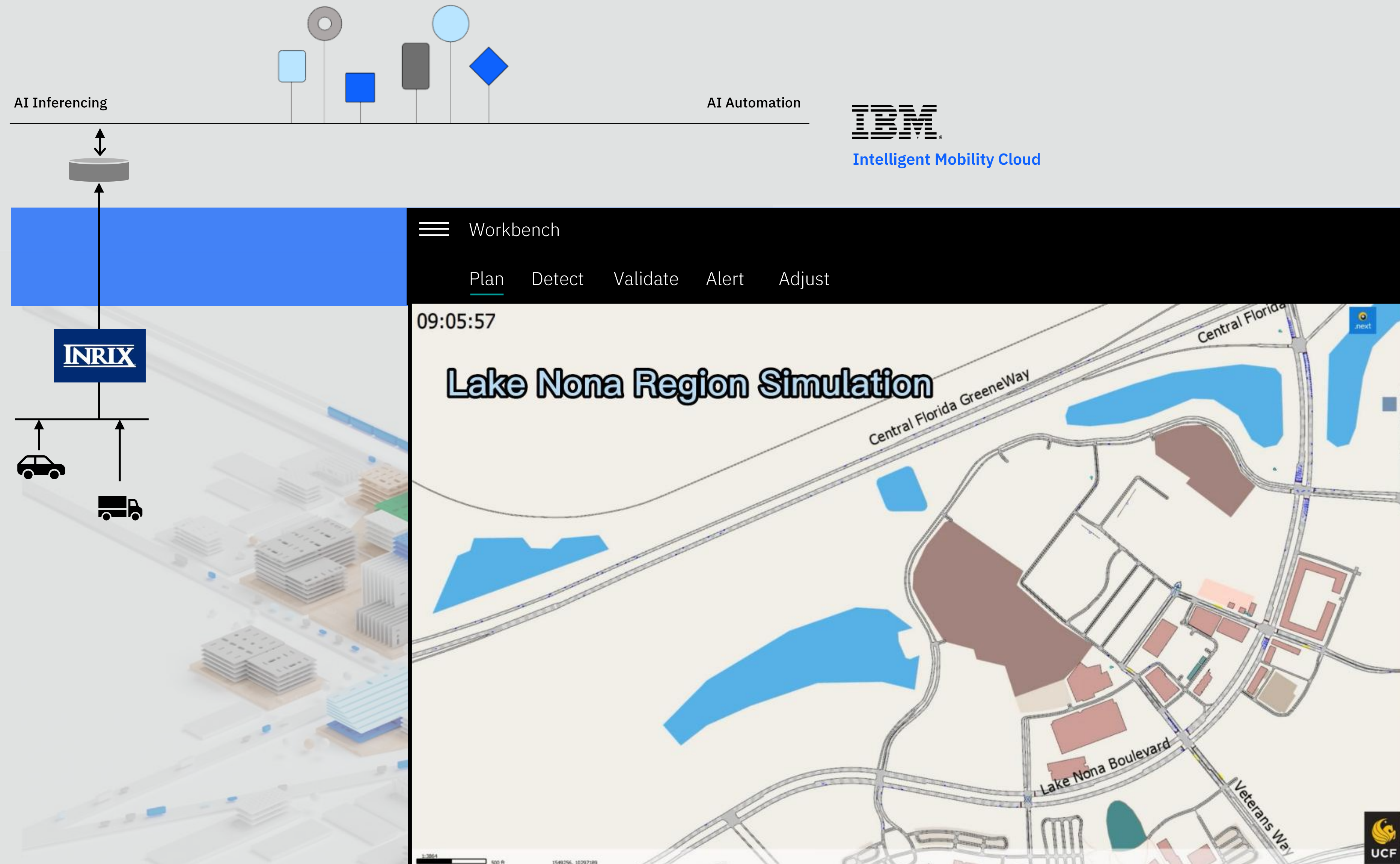


Historical traffic congestion using probe vehicle data presented by time and location, integrated from INRIX

Traffic

Demand sensing can be used to adjust demand fulfillment.

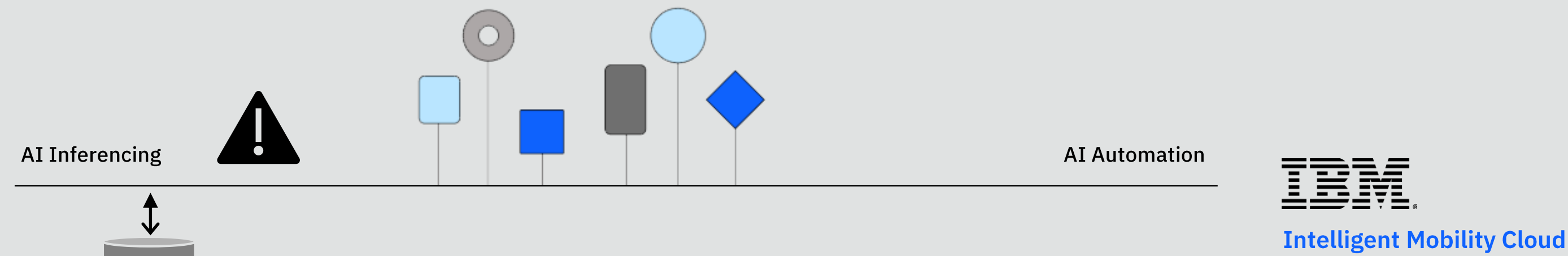
1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume



Traffic

Demand sensing can be used to adjust demand fulfillment.

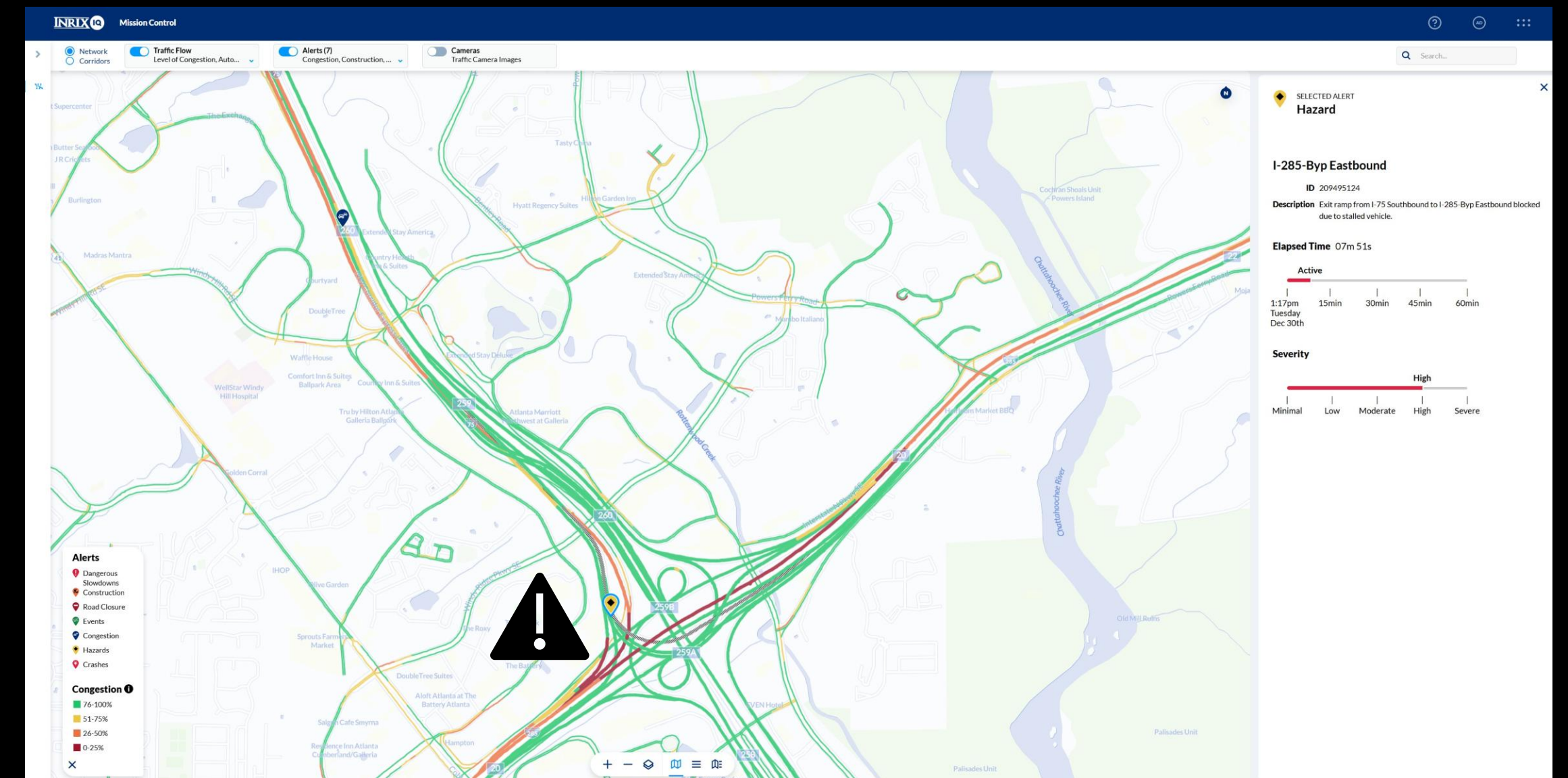
1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume



IBM
Intelligent Mobility Cloud

Workbench

Plan Detect Validate Alert Adjust

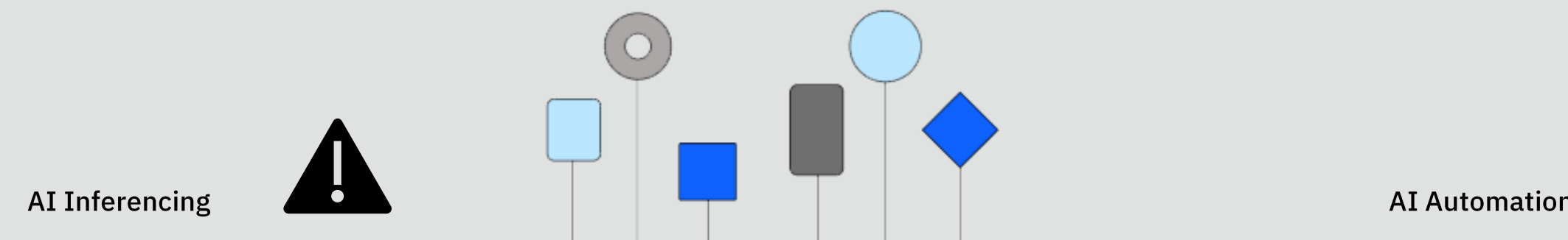


Real-time congestion detection by INRIX

Traffic

Demand sensing can be used to adjust demand fulfillment.

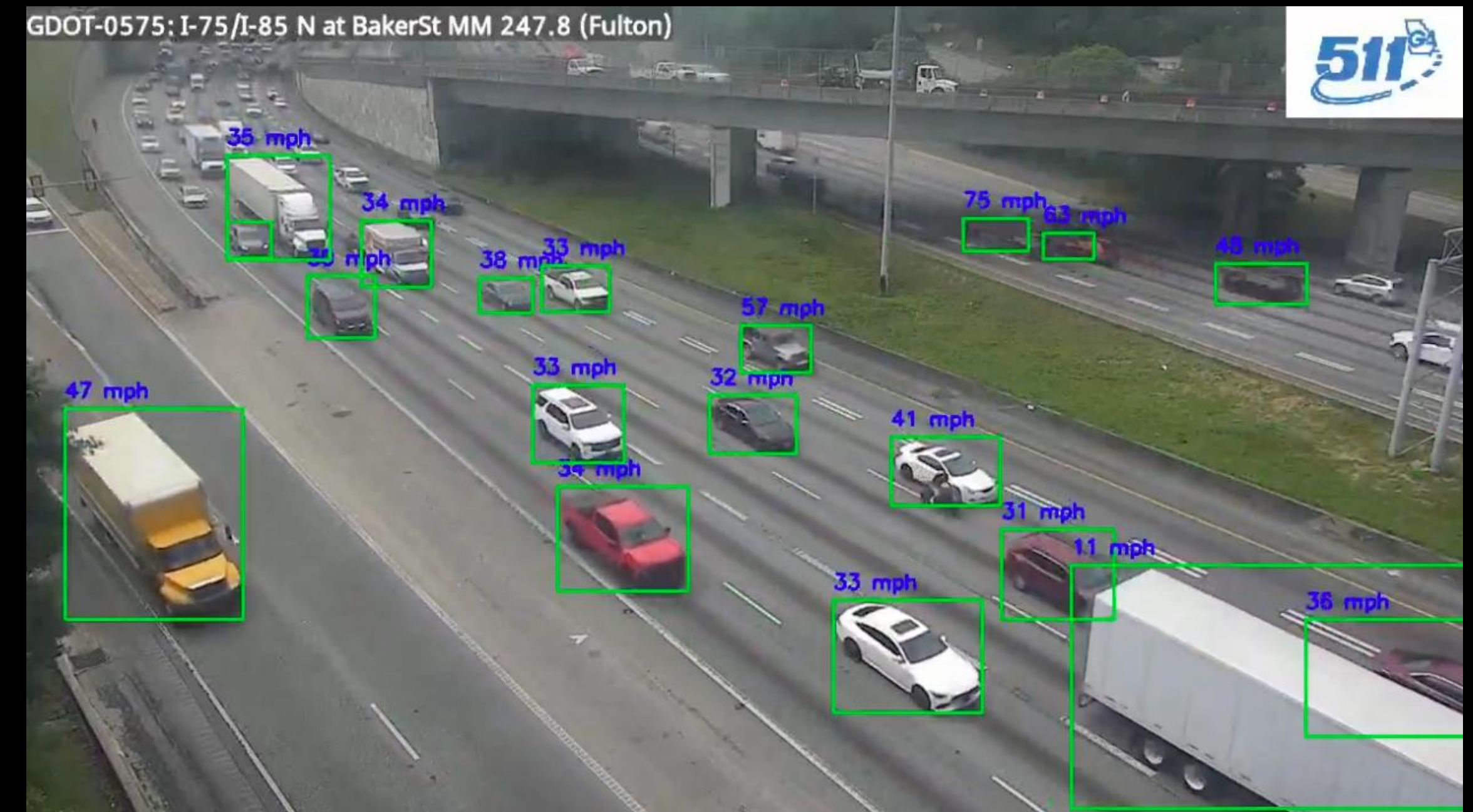
1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume



IBM
Intelligent Mobility Cloud

Workbench

Plan Detect Validate Alert Adjust



Visual analytics integrated from egen using available GDOT camera feeds, to validate and enhance probe vehicle indications

Traffic

Demand sensing can be used to adjust demand fulfillment.

1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume

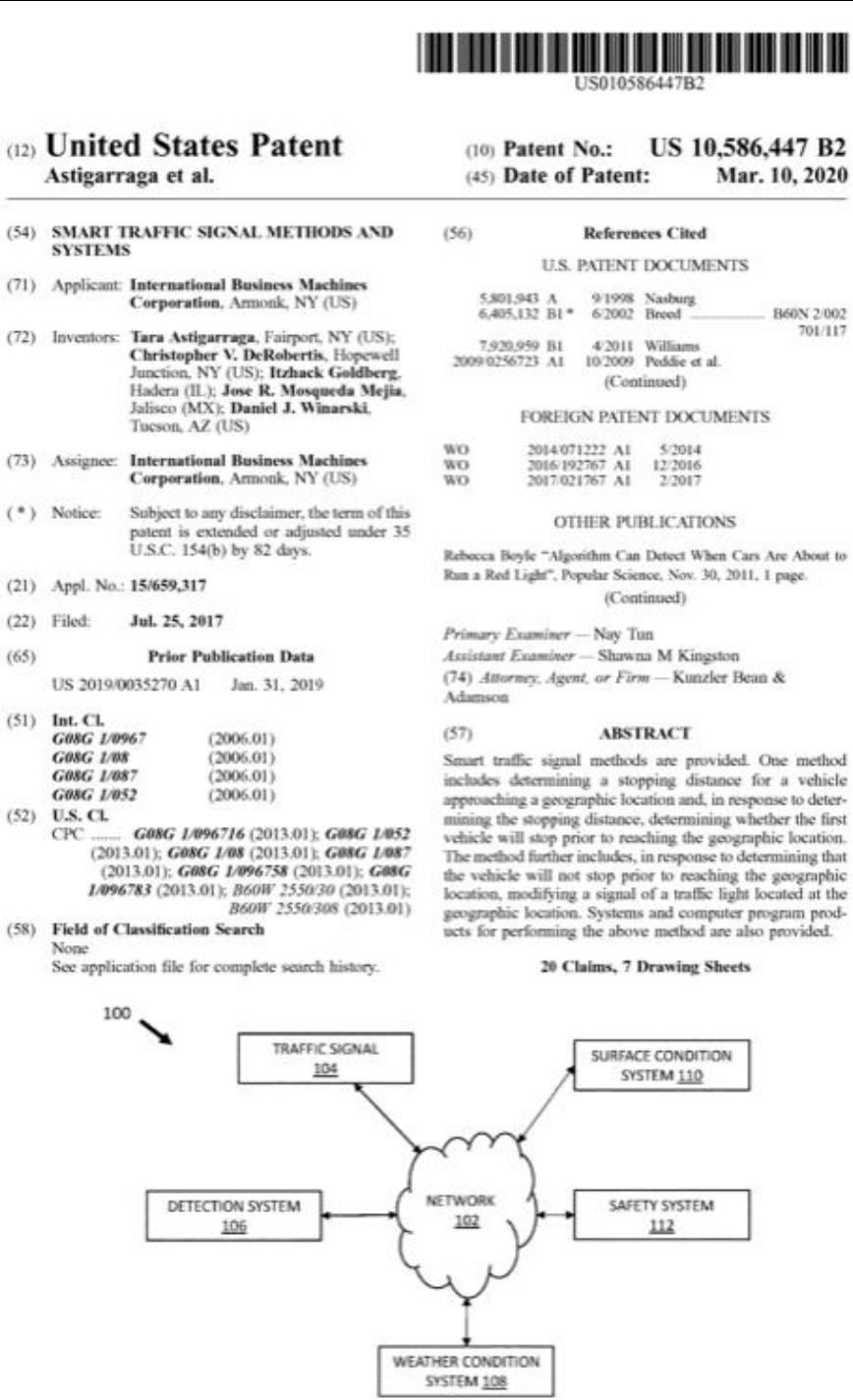
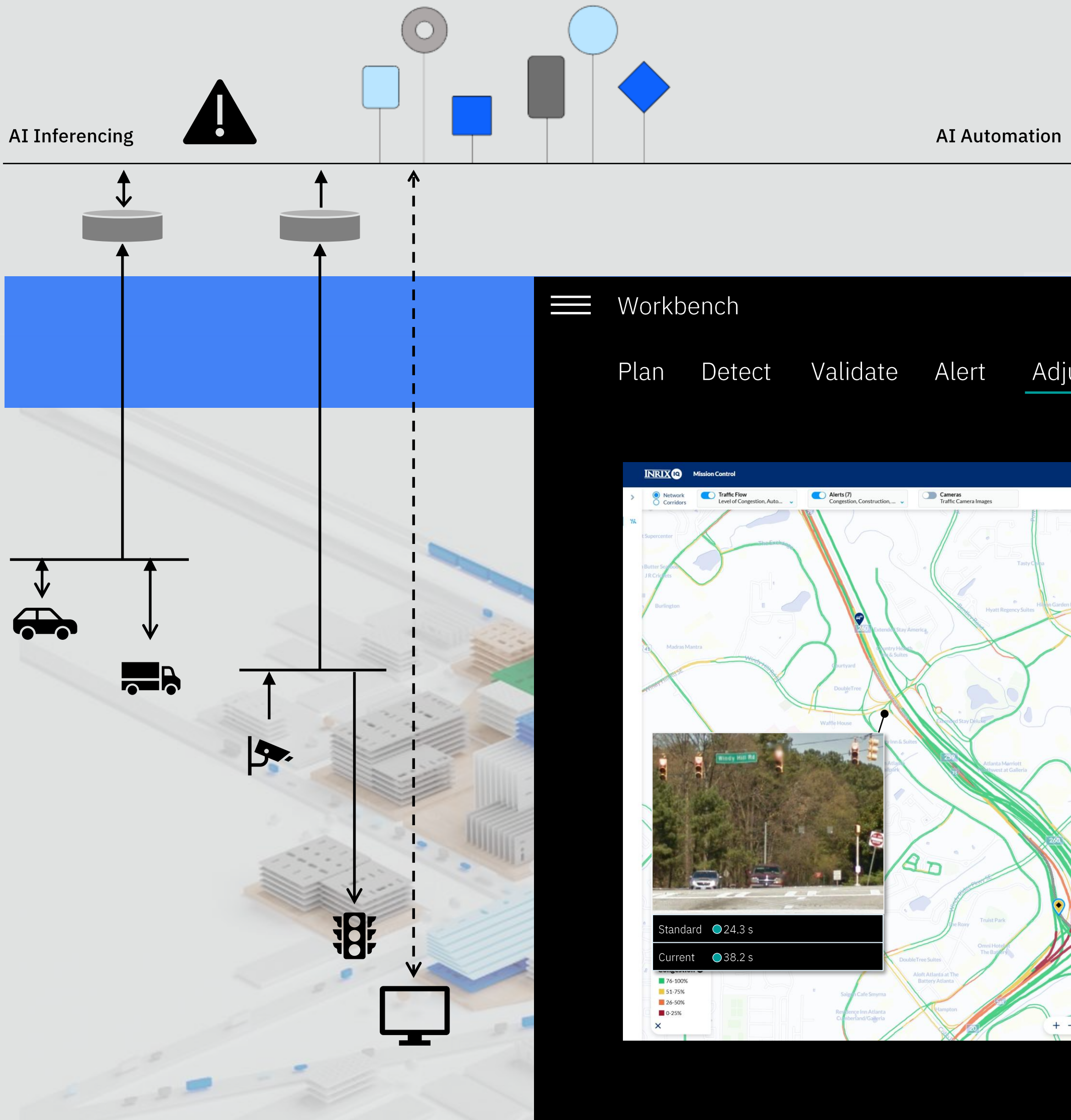


Push of notifications via IBM CVI to participating auto OEMs for supported vehicles. Honda model shown using Android auto already onboard.

Traffic

Demand sensing can be used to adjust demand fulfillment.

- 1. **Plan:** Baseline playbooks are tuned based on probe vehicle data
- 2. **Detect:** Data from connected probe vehicles is consumed by INRIX, and detects increased congestion on the highway and the feeder road network
- 3. **Validate:** Alerts are validated automatically using visual analytics from the camera system already installed
- 4. **Alert:** Messages pushed to cars and signs on highway and feeder roads about congestion and to use Express Lanes
- 5. **Adjust:** Traffic lights at the onramp junctions can be adjusted dynamically to throttle flow into the highway until capacity can handle additional volume



Signal optimization using coordinated patterns – red lights upstream from the congestion to reduce flow, and green lights downstream to flush the queue.



Thank you

Donovan Guin

Director, Public
Transportation

dguin@ibm.com

